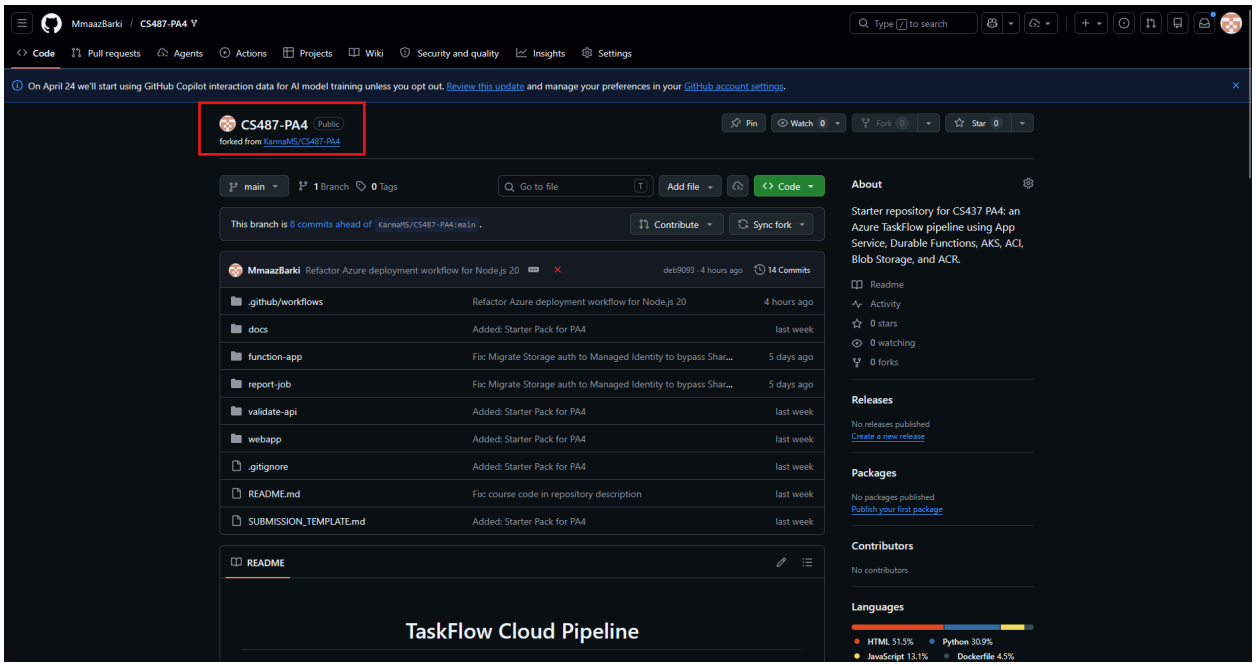
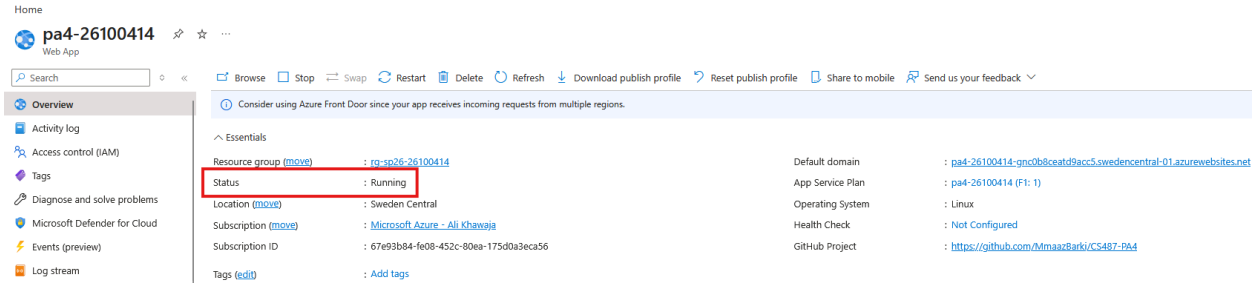
Task 1:

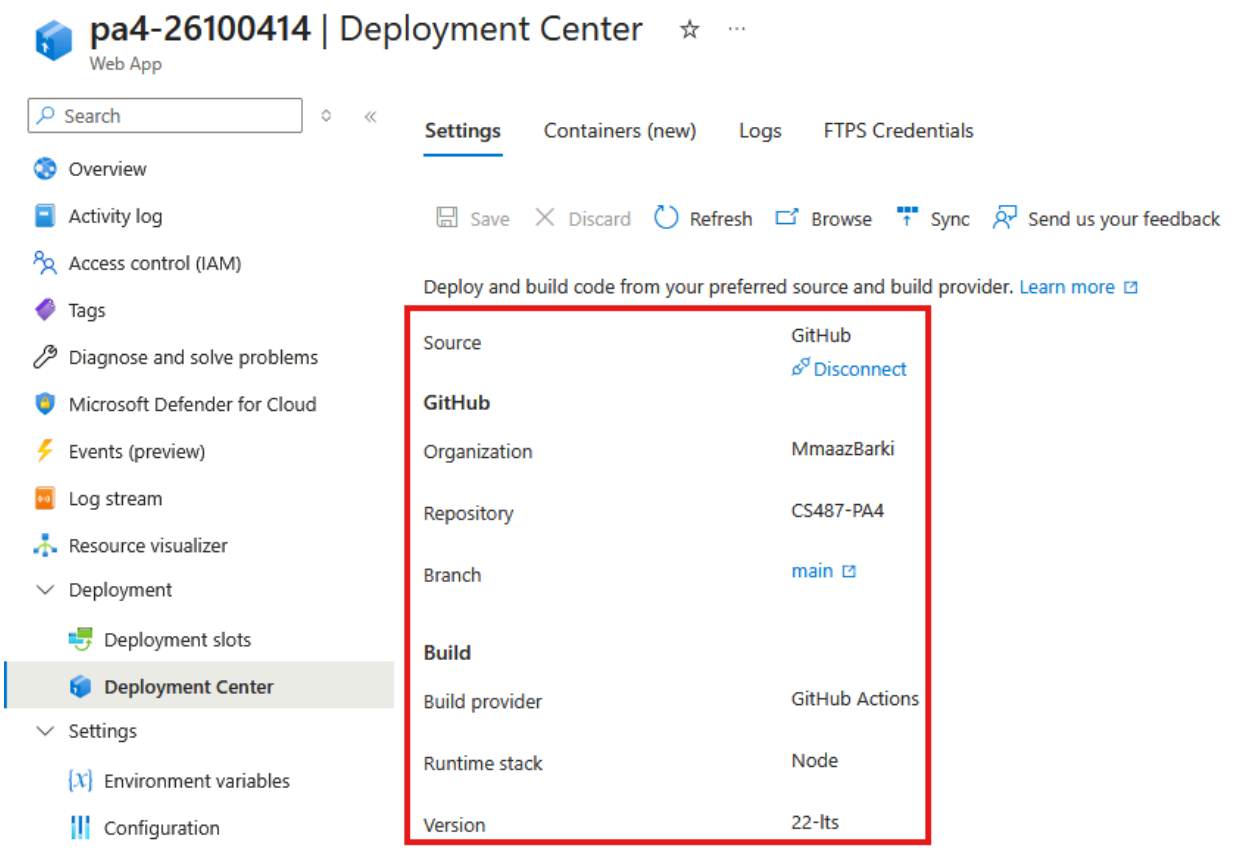
Forked GitHub Repo



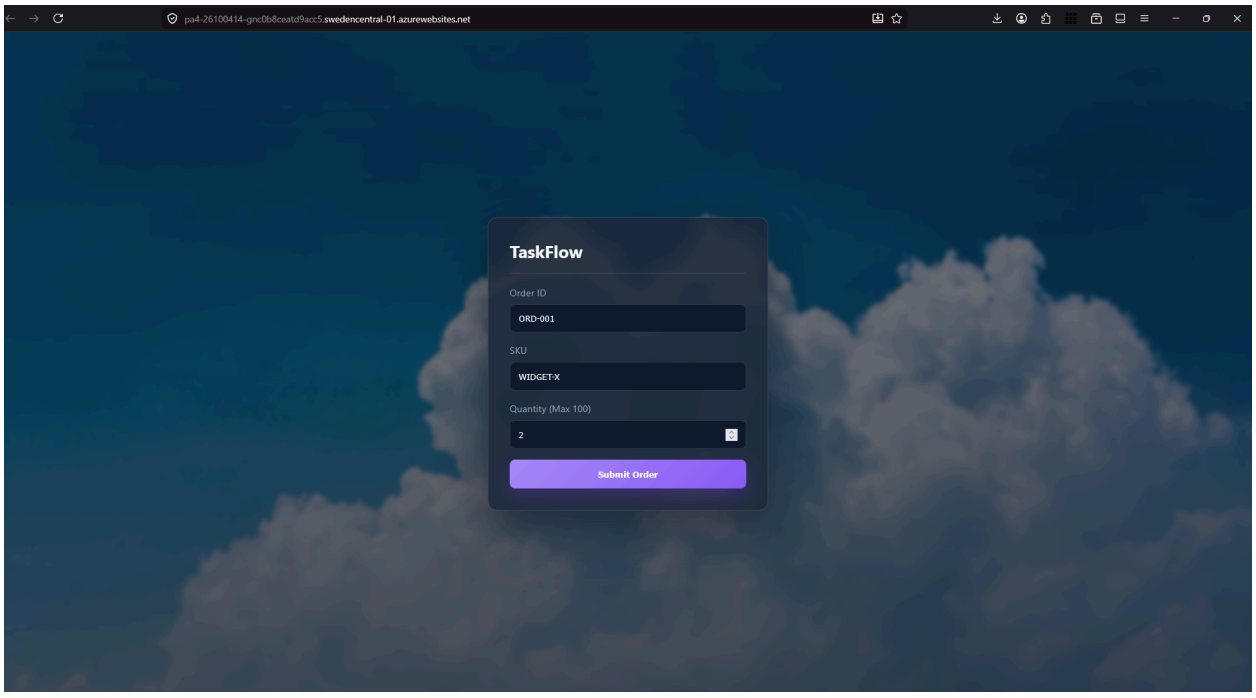
Web App Overview: Running Status



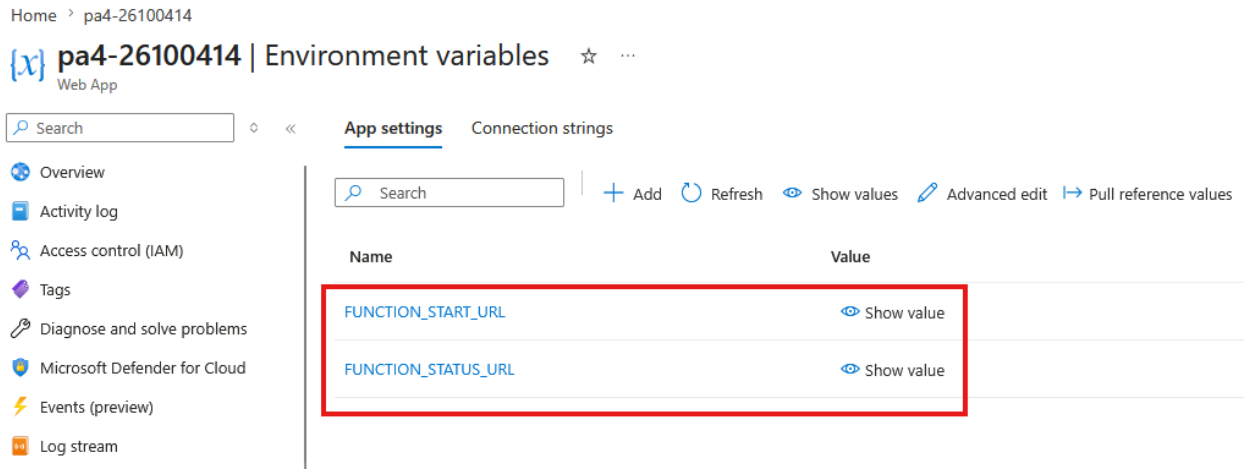
Deployment Center: GitHub Source



Dashboard Loading in Browser:

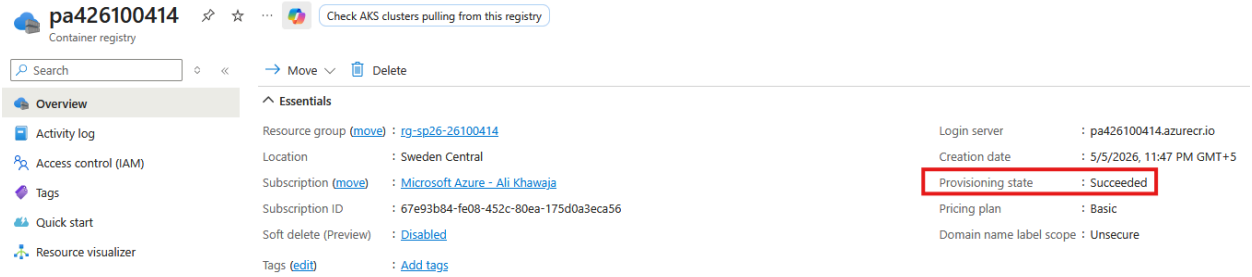


Configured Application Settings:



Task 2:

ACR Overview Success:



Successful Docker Builds

```
(base) PS C:\Users\maazb\OneDrive\Desktop\Cloud-Dev\Assignments\Assignment4\CS487-PA4> docker build -t validate-api:latest ./validate-api
[+] Building 65.6s (11/11) FINISHED                                                                    docker:desktop-linux
```

```
(base) PS C:\Users\maazb\OneDrive\Desktop\Cloud-Dev\Assignments\Assignment4\CS487-PA4> docker build -t report-job:latest ./report-job
[+] Building 17.7s (10/10) FINISHED                                                                    docker:desktop-linux
```

```
(base) PS C:\Users\maazb\OneDrive\Desktop\Cloud-Dev\Assignments\Assignment4\CS487-PA4> docker build -t func-app:latest ./function-app
[+] Building 2399.9s (9/9) FINISHED                                                                    docker:desktop-linux
```

Successful local validator test

```
(base) PS C:\Users\maazb\OneDrive\Desktop\Cloud-Dev\Assignments\Assignment4\CS487-PA4> curl.exe -X POST http://localhost
:8080/validate -H "Content-Type: application/json" -d "{\'"order_id\'":\'"LOCAL-1\'",\'"items\'":[{\'"sku\'":\'"ABC\'",\
{"valid":true,"reason":"ok","order_id":"LOCAL-1"}
```

Image Tagging Commands:
docker tag validate-api:latest pa426100414.azurecr.io/validate-api:v1
docker tag report-job:latest pa426100414.azurecr.io/report-job:v1
docker tag func-app:latest pa426100414.azurecr.io/func-app:v1

Image Pushing Commands:
docker push pa426100414.azurecr.io/validate-api:v1
docker push pa426100414.azurecr.io/report-job:v1
docker push pa426100414.azurecr.io/func-app:v1

Successful Image Push:

```
(base) PS C:\Users\maazb\OneDrive\Desktop\Cloud-Dev\Assignments\Assignment4\CS487-PA4> docker push pa426100414.azurecr.io/validate-api:v1
The push refers to repository [pa426100414.azurecr.io/validate-api]
a17f4a15dff7: Pushed
7ccd73948dde: Pushed
91ff8760033c: Pushed
3531af2bc2a9: Pushed
de43e94a3438: Pushed
f3ba2250c524: Pushed
d915f35567d3: Pushed
be7084d609f6: Pushed
994b958a9456: Pushed
v1: digest: sha256:0f194559a650670e552404572dfeacec05296ac2750c395633afb15643bbae11 size: 856
```

```
(base) PS C:\Users\maazb\OneDrive\Desktop\Cloud-Dev\Assignments\Assignment4\CS487-PA4> docker push pa426100414.azurecr.io/report-job:v1
The push refers to repository [pa426100414.azurecr.io/report-job]
3d855105a4a0: Pushed
f3ba2250c524: Mounted from validate-api
91ff8760033c: Mounted from validate-api
7ccd73948dde: Mounted from validate-api
de43e94a3438: Mounted from validate-api
22d84cb0741a: Pushed
934d1e46c047: Pushed
3531af2bc2a9: Mounted from validate-api
93021b56d2d2: Pushed
v1: digest: sha256:27e4ad223c58771ae560878ff53686b20c53e7f4da2e62ff1e0a62d76862615f size: 856
```

```
(base) PS C:\Users\maazb\OneDrive\Desktop\Cloud-Dev\Assignments\Assignment4\CS487-PA4> docker push pa426100414.azurecr.io/func-app:v1
The push refers to repository [pa426100414.azurecr.io/func-app]
8b5c9ed400b7: Pushed
4f4fb700ef54: Pushed
af5ac245ac18: Pushed
48805a8c2422: Pushed
265a95973510: Pushed
295675d49bd5: Pushed
cc2ef16d9998: Pushed
2cff7b2fc77f: Pushed
1e291ff5c47c: Pushed
6d411a239a49: Pushed
c2b164019d21: Pushed
83dc37193a82: Pushed
5999d6a7b922: Pushed
66f110fb4b1b: Pushed
8715c03abd5b: Pushed
f6e58f04b6b2: Pushed
40104569dda3: Pushed
60993470af6a: Pushed
3263c6727b31: Pushed
9e50f0fc8e7b: Pushed
3b3815156222: Pushed
89320e7119e2: Pushed
v1: digest: sha256:958dfe01a60bffc3624b8ed19c1f6b46adf5daadecdd379d5162e16be9349276 size: 856
```

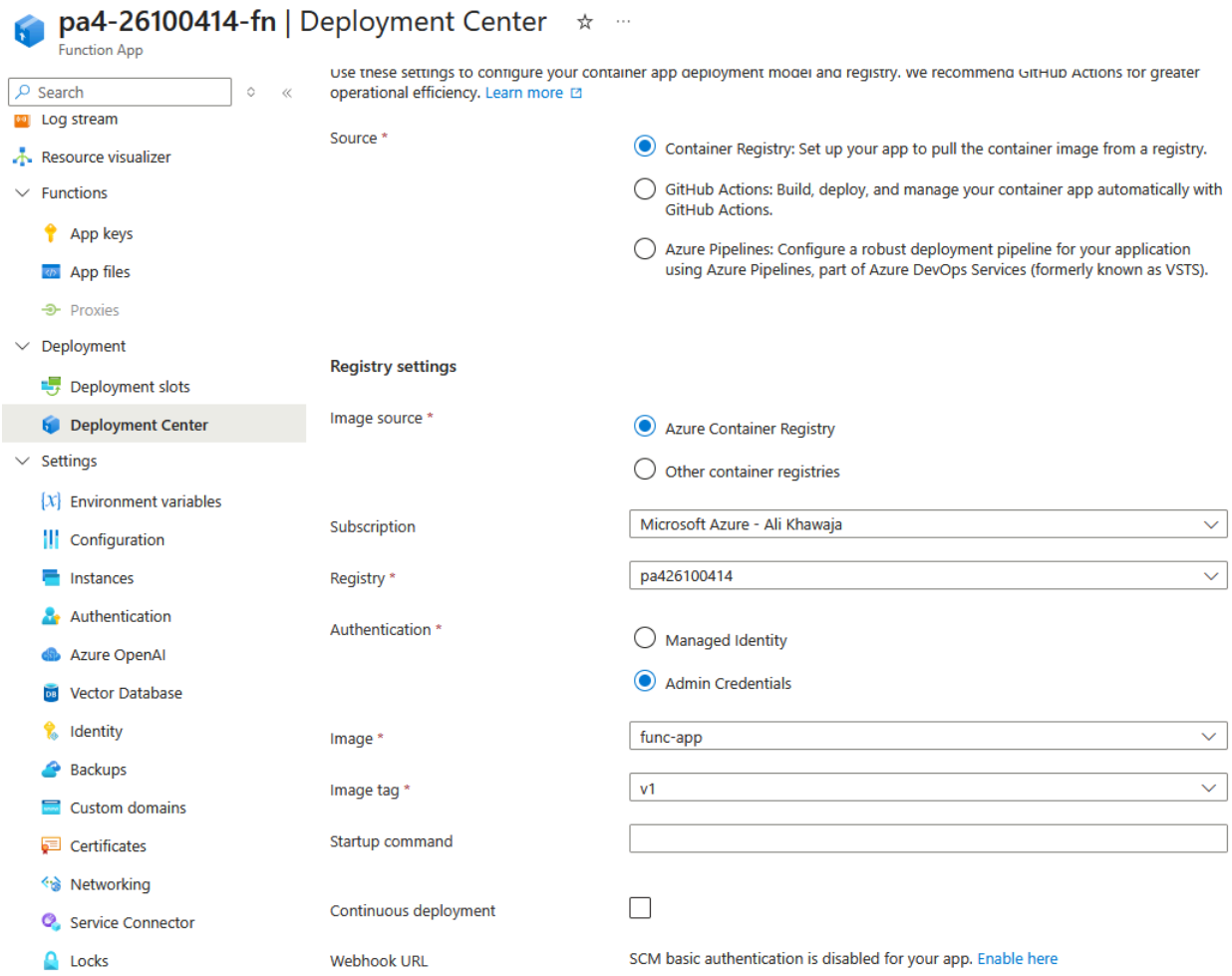
Task 3:

Durable Functions visible on func start:

```
(.venv) (base) PS C:\Users\maazb\OneDrive\Desktop\Cloud-Dev\Assignments\Assignment4\CS487-PA4\function-app> func start


Azure Functions Core Tools
Core Tools Version:       4.10.0+2ced9a89e29b04615bc55cd527c074250266362f (64-bit)
Function Runtime Version: 4.1048.200.26180

[2026-05-06T07:39:45.132Z] Durable Functions Distributed Tracing V2 is GA now! Learn how to enable the feature by visiti
ng aka.ms/durable-distributed-tracing. To disable this message, you can configure distributedTracingEnabled to "true" an
d version to "V2" or "None". Setting it to "None" would in effect disable the feature.
[2026-05-06T07:39:45.432Z] Worker process started and initialized.

Functions:

        http_starter: [POST] http://localhost:7071/api/orchestrators/my_orchestrator

        my_orchestrator: orchestrationTrigger

        report_activity: activityTrigger

        validate_activity: activityTrigger

For detailed output, run func with --verbose flag.
```

Task 4:

Container Image Configuration:



Functions list in Overview:

Name	Trigger	Status	Monitor
http_starter	HTTP	Enabled	Invocations and more
my_orchestrator	Orchestration	Enabled	Orchestration information
report_activity	Activity	Enabled	Invocations and more
validate_activity	Activity	Enabled	Invocations and more

Curl output:

```
maaz [ ~ ]$ curl -X POST "https://pa4-26100414-fn-a5ereuczf9cue4cj.swedencentral-01.azurewebsites.net/api/orchestrators/my_orchestrator?code=88y_Cdax_b7q3Fq6XDRxZNUL3dCaV3y9Hdc8K8hE0uOLAzFuozBNzw==" -H "Content-Type: application/js
on" -d '{"order_id": "FINAL-SMOKE-TEST", "items": [{"sku": "ABC", "qty": 1}]}'
{"id": "50a12e51df4e4bc795e268d67f5a8a43", "statusQueryGetUri": "https://pa4-26100414-fn-a5ereuczf9cue4cj.swedencentral-01.azurewebsites.net/runtime/webhooks/durabletask/instances/50a12e51df4e4bc795e268d67f5a8a43?taskHub=pa426100414fn&connection=Storage&code=YMHgkSKvnrXVBxVmF1KXzD5W6SPfHyansFxgT4evhLUUAzFucmpr9g==", "sendEventPostUri": "https://pa4-26100414-fn-a5ereuczf9cue4cj.swedencentral-01.azurewebsites.net/runtime/webhooks/durabletask/instances/50a12e51df4e4bc795e268d67f5a8a43/raiseEvent/{eventName}?taskHub=pa426100414fn&connection=Storage&code=YMHgkSKvnrXVBxVmF1KXzD5W6SPfHyansFxgT4evhLUUAzFucmpr9g==", "terminatePostUri": "https://pa4-26100414-fn-a5ereuczf9cue4cj.swedencentral-01.azurewebsites.net/runtime/webhooks/durabletask/instances/50a12e51df4e4bc795e268d67f5a8a43/terminate?reason={text}&taskHub=pa426100414fn&connection=Storage&code=YMHgkSKvnrXVBxVmF1KXzD5W6SPfHyansFxgT4evhLUUAzFucmpr9g==", "rewindPostUri": "https://pa4-26100414-fn-a5ereuczf9cue4cj.swedencentral-01.azurewebsites.net/runtime/webhooks/durabletask/instances/50a12e51df4e4bc795e268d67f5a8a43/rewind?reason={text}&taskHub=pa426100414fn&connection=Storage&code=YMHgkSKvnrXVBxVmF1KXzD5W6SPfHyansFxgT4evhLUUAzFucmpr9g==", "purgeHistoryDeleteUri": "https://pa4-26100414-fn-a5ereuczf9cue4cj.swedencentral-01.azurewebsites.net/runtime/webhooks/durabletask/instances/50a12e51df4e4bc795e268d67f5a8a43?taskHub=pa426100414fn&connection=Storage&code=YMHgkSKvnrXVBxVmF1KXzD5W6SPfHyansFxgT4evhLUUAzFucmpr9g==", "restartPostUri": "https://pa4-26100414-fn-a5ereuczf9cue4cj.swedencentral-01.azurewebsites.net/runtime/webhooks/durabletask/instances/50a12e51df4e4bc795e268d67f5a8a43/restart?taskHub=pa426100414fn&connection=Storage&code=YMHgkSKvnrXVBxVmF1KXzD5W6SPfHyansFxgT4evhLUUAzFucmpr9g==", "suspendPostUri": "https://pa4-26100414-fn-a5ereuczf9cue4cj.swedencentral-01.azurewebsites.net/runtime/webhooks/durabletask/instances/50a12e51df4e4bc795e268d67f5a8a43/suspend?reason={text}&taskHub=pa426100414fn&connection=Storage&code=YMHgkSKvnrXVBxVmF1KXzD5W6SPfHyansFxgT4evhLUUAzFucmpr9g==", "resumePostUri": "https://pa4-26100414-fn-a5ereuczf9cue4cj.swedencentral-01.azurewebsites.net/runtime/webhooks/durabletask/instances/50a12e51df4e4bc795e268d67f5a8a43/resume?reason={text}&taskHub=pa426100414fn&connection=Storage&code=YMHgkSKvnrXVBxVmF1KXzD5W6SPfHyansFxgT4ev
```

"statusQueryGetUri":
"https://pa4-26100414-fn-a5ereuczf9cue4cj.swedencentral-01.azurewebsites.net/runtime/webhooks/durabletask/instances/50a12e51df4e4bc795e268d67f5a8a43?taskHub=pa426100414fn&connection=Storage&code=YMHgkSKvnrXVBxVmF1KXzD5W6SPfHyansFxgT4evhLUUAzFucmpr9g=="
"Id":
"50a12e51df4e4bc795e268d67f5a8a43"

Status URL Query Json:

```
name:              "my_orchestrator"
instanceId:        "50a12e51df4e4bc795e268d67f5a8a43"
runtimeStatus:     "Completed"
input:             '{"order_id": "FINAL-SMOKE-TEST", "items": [{"sku": "ABC", "qty": 1}]}'
customStatus:      null
output:            null
createdTime:       "2026-05-06T18:46:09Z"
lastUpdatedTime:   "2026-05-06T18:46:10Z"
```

Task 5

Kubectl get nodes:

```
(base) PS C:\Users\maazb\OneDrive\Desktop\Cloud-Dev\Assignments\Assignment4\CS487-PA4> kubectl get nodes
NAME                                STATUS   ROLES    AGE   VERSION
aks-nodepool1-34692032-vmss000000   Ready    <none>   17m   v1.34.6
```

Kubectl get pods:

```
(base) PS C:\Users\maazb\OneDrive\Desktop\Cloud-Dev\Assignments\Assignment4\CS487-PA4> kubectl get pods
NAME                                  READY   STATUS    RESTARTS   AGE
validate-deployment-577f889bd7-d65mj  1/1     Running   0          14m
```

Kubectl get service:

```
(base) PS C:\Users\maazb\OneDrive\Desktop\Cloud-Dev\Assignments\Assignment4\CS487-PA4> kubectl get service
NAME               TYPE           CLUSTER-IP     EXTERNAL-IP   PORT(S)          AGE
kubernetes         ClusterIP      10.0.0.1       <none>        443/TCP          20m
validate-service   LoadBalancer   10.0.217.194   20.90.53.87   8080:32749/TCP   15m
```

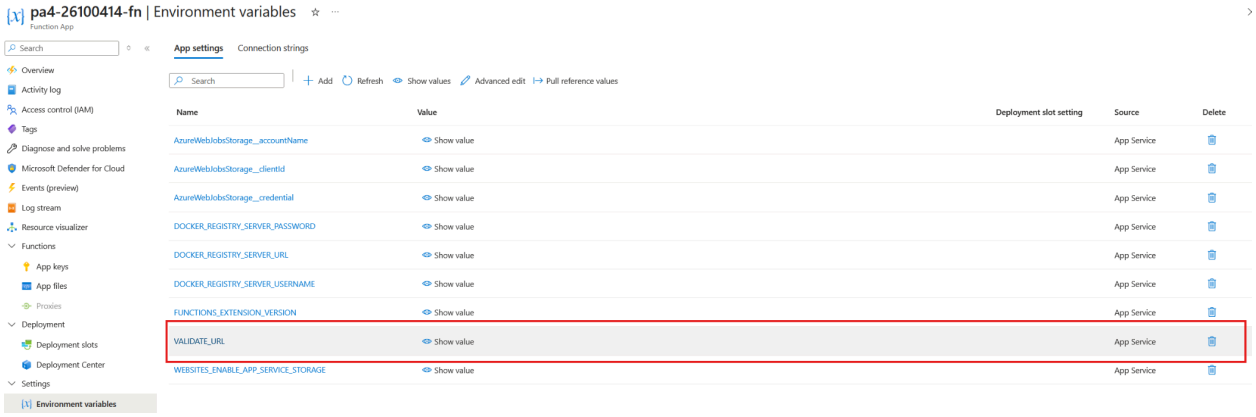
Good order:

```
(base) PS C:\Users\maazb\OneDrive\Desktop\Cloud-Dev\Assignments\Assignment4\CS487-PA4> Invoke-RestMethod -Uri "http://20
.90.53.87:8080/validate" `
>>   -Method Post `
>>   -ContentType "application/json" `
>>   -Body '{"order_id":"O-1001","items":[{"sku":"ABC","qty":2}]}'

valid reason order_id
----- ------ --------
 True ok     O-1001
```

Bad order:

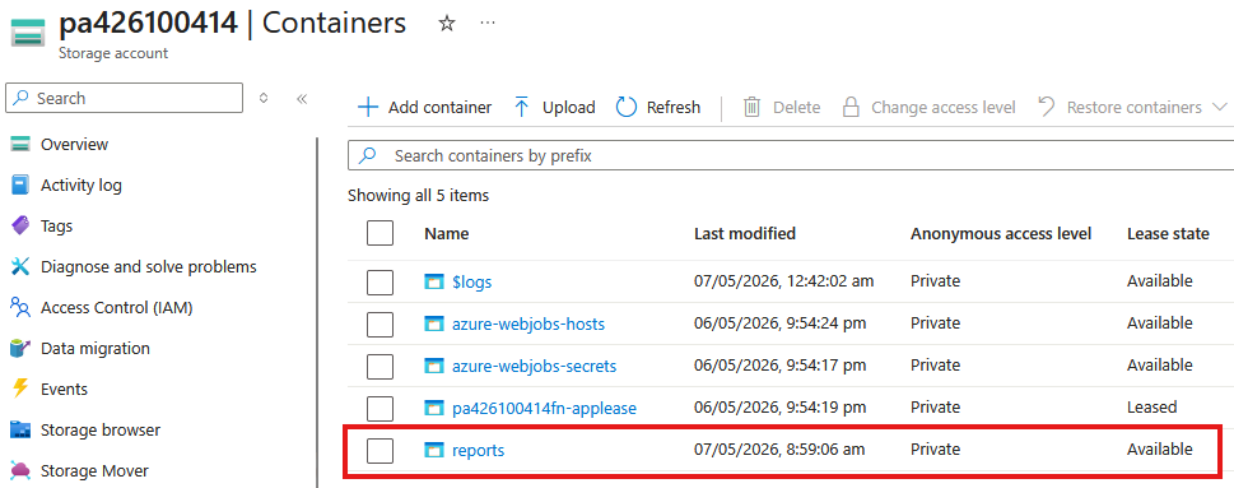
```
(base) PS C:\Users\maazb\OneDrive\Desktop\Cloud-Dev\Assignments\Assignment4\CS487-PA4> Invoke-RestMethod -Uri "http://20
.90.53.87:8080/validate" `
>>   -Method Post `
>>   -ContentType "application/json" `
>>   -Body '{"order_id":"O-1002","items":[{"sku":"ABC","qty":999}]}'

valid reason                 order_id
----- ------                 --------
False quantity exceeds limit O-1002
```

Function App Application Setting (Validate URL):



Task 6

reports blob creation:



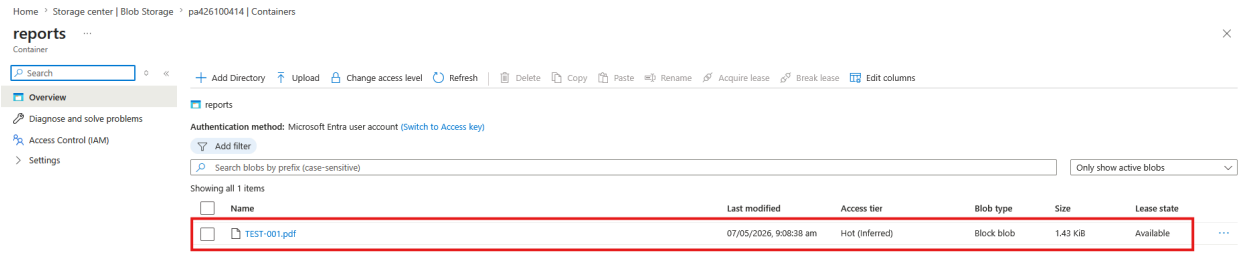
Successful container show (for **ci-report-test**):

```
maaz [ ~ ]$ az container show --resource-group rg-sp26-26100414 --name ci-report-test --query "instanceView.state" --output tsv
Succeeded
```

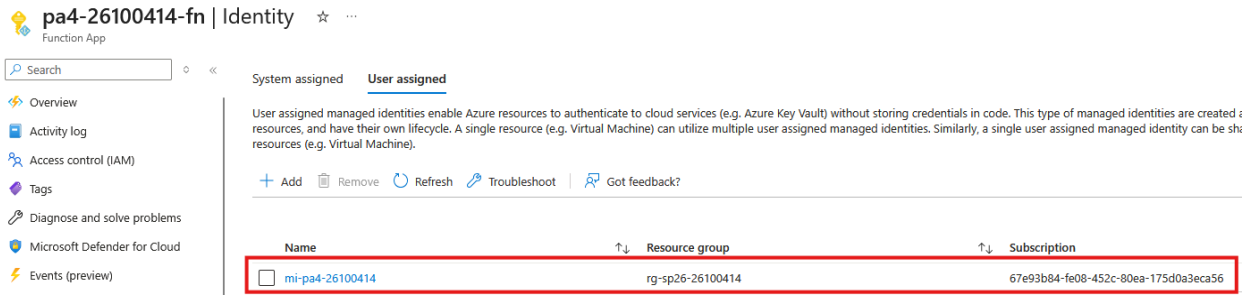
Container logs terminal output:

```
maaz [ ~ ]$ az container logs --resource-group rg-sp26-26100414 --name ci-report-test
Uploaded TEST-001.pdf to reports container
```

PDF in reports container:



Function App Identity Blade (showing Managed Identity):



Function App settings

REPORT_*:

REPORT_IMAGE	Show value	App Service	
REPORT_LOCATION	Show value	App Service	
REPORT_RG	Show value	App Service	

ACR_*:

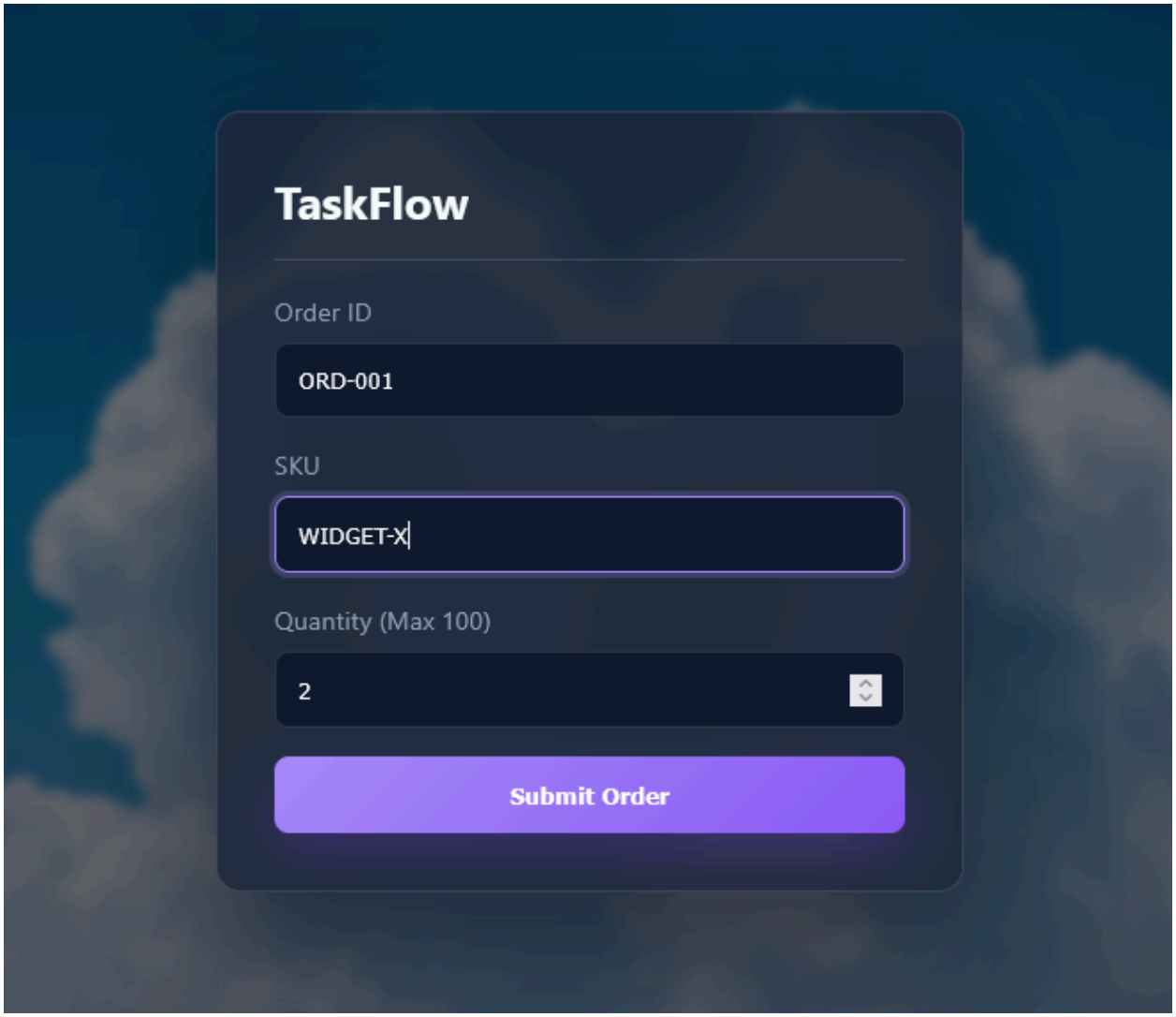
ACR_PASSWORD	Show value	App Service	
ACR_SERVER	Show value	App Service	
ACR_USERNAME	Show value	App Service	

STORAGE_ACCOUNT_URL:

STORAGE_ACCOUNT_URL	Show value	App Service	

Task 7

Filled Form:



Function Invocations (Log Stream):

```
maaz [ ~ ]$ az webapp log tail --name pa4-26100414-fn --resource-group rg-sp26-26100414
2026-05-07T15:15:26  Welcome, you are now connected to log-streaming service.
Starting Log Tail -n 10 of existing logs ----
/appsvctmp/volatile/logs/runtime/container.log
2026-05-07T14:54:36.0640247Z info: Function.my_orchestrator[2]
2026-05-07T14:54:36.0640506Z       Executed 'Functions.my_orchestrator' (Succeeded, Id=432efd2e-a925-43fd-90f7-926b812b0b56, Duration=37ms)
2026-05-07T14:54:36.0646806Z MS_FUNCTION_AZURE_MONITOR_EVENT 4,pa4-26100414-fn-a5ereuczf9cue4cj.swedencentral-01.azurewebsites.net,Microsoft.Web/sites/functions/log,Functi
onAppLogs,swedencentral,"{'appName':'pa4-26100414-fn','roleInstance':'9d4c7e962a9486d3351a27c2a032aad96892fc125f45ab5254d12169dae7fd2f','message':'Executed Functions.my_or
chestrator (Succeeded, Id=432efd2e-a925-43fd-90f7-926b812b0b56, Duration=37ms)','category':'Function.my_orchestrator','hostVersion':'4.1048.200.5','functionInvocationId':'
432efd2e-a925-43fd-90f7-926b812b0b56','functionName':'Functions.my_orchestrator','hostInstanceId':'4c4d2f61-e59e-4aed-869c-1611d839274f','level':'Information','levelId':2,
'processId':9,'eventId':2,'eventName':'FunctionCompleted'}",05/07/2026 14:54:36
2026-05-07T14:54:39.3156908Z info: Microsoft.Azure.WebJobs.Hosting.OptionsLoggingService[0]
2026-05-07T14:54:39.3157179Z       ExtensionSystemOptions
2026-05-07T14:54:39.3157201Z       {
2026-05-07T14:54:39.3157216Z         "extensionName": "durabletask",
2026-05-07T14:54:39.3157231Z         "webhookAuthorizationLevel": "System"
2026-05-07T14:54:39.3157277Z       }
2026-05-07T14:54:39.3230173Z MS_FUNCTION_AZURE_MONITOR_EVENT 4,pa4-26100414-fn-a5ereuczf9cue4cj.swedencentral-01.azurewebsites.net,Microsoft.Web/sites/functions/log,Functi
onAppLogs,swedencentral,"{'appName':'pa4-26100414-fn','roleInstance':'9d4c7e962a9486d3351a27c2a032aad96892fc125f45ab5254d12169dae7fd2f','message':'ExtensionSystemOptions\n
{\n  \'extensionName\': \'durabletask\',\n  \'webhookAuthorizationLevel\': \'System\'\n}','category':'Microsoft.Azure.WebJobs.Hosting.OptionsLoggingService','hostVersion':
'4.1048.200.5','hostInstanceId':'4c4d2f61-e59e-4aed-869c-1611d839274f','level':'Information','levelId':2,'processId':9}",05/07/2026 14:54:39
Ending Log Tail of existing logs ---
Starting Live Log Stream ---
2026-05-07T15:16:26  No new trace in the past 1 min(s).
```